
CONCORDIUM

Introduction to Concordium

Presenters:

Dragos Danciulescu - Developer Relations Engineer
Doris Benda - Senior Software Engineer

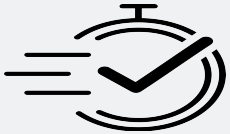
Useful Links

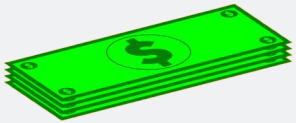
- Follow Along, access links, presentation link
- Official Concordium Documentation
- Official Concordium Website
- Concordium Github
- Join our Discord!
- Whitepaper

Facts and Stats

Facts and Stats

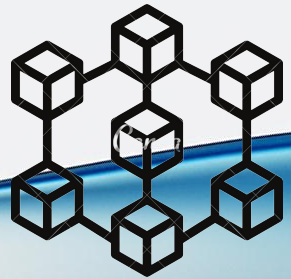
- Up to 2000 TPS (scalable beyond this threshold)
- 2-4 seconds Transaction Finality
- Low transaction fee (€0.01), pegged to the Euro;
- 172,999,681 total transactions currently
- 103,173 accounts currently
- 9956 smart contracts deployed
- 11 stablecoins as Protocol-Level Tokens
- Concordium BFT (Byzantine Fault Tolerance) consensus protocol
- Open-source project











Blockchain + Identity = Concordium

- **Identity**: Requires ID check to open Accounts
- ZK-Proofs: Verify information about real-world identities
- **Public**: Anyone can run a node

- **Support for multiple programming languages**, main ones are Rust and Typescript
- **Science-based**: Level 1 blockchain built by leading scientists
- **Proof-of-stake**: Eco-friendly. Weighted lottery tokenomics

Principles of Privacy

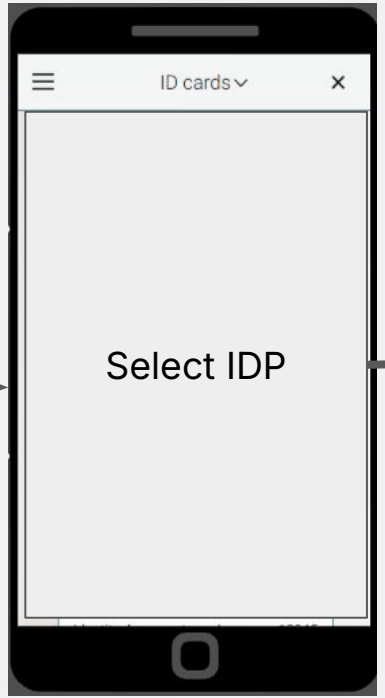
Concordium’s identity system has been designed to be privacy first and incorporates several key privacy principles.

Principles of Privacy - Protection of PII

Users' **personal identifiable information (PII)** such as names, birth dates etc, is never available on chain, either encrypted or unencrypted. PII is only stored locally within the user's wallet and with the Identity Provider (IDP) for compliance purposes—users retain full control and ownership of their data at all times. Concordium does not have access to, own, or store any of this user information.

Account Opening Flow

YOU

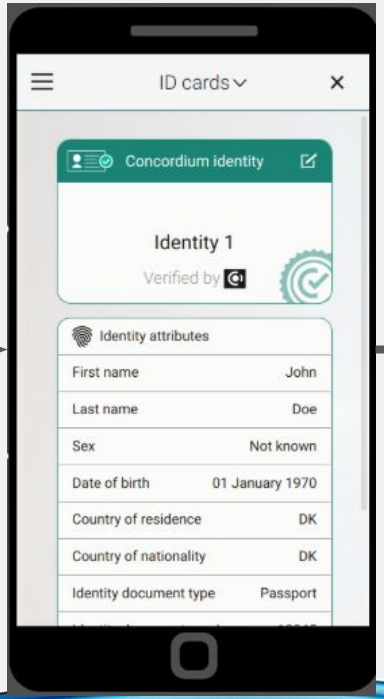


User Wallet

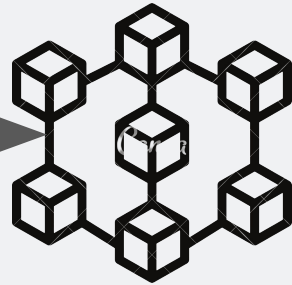


IDP checks ID and liveness

Identity Provider (IDP)



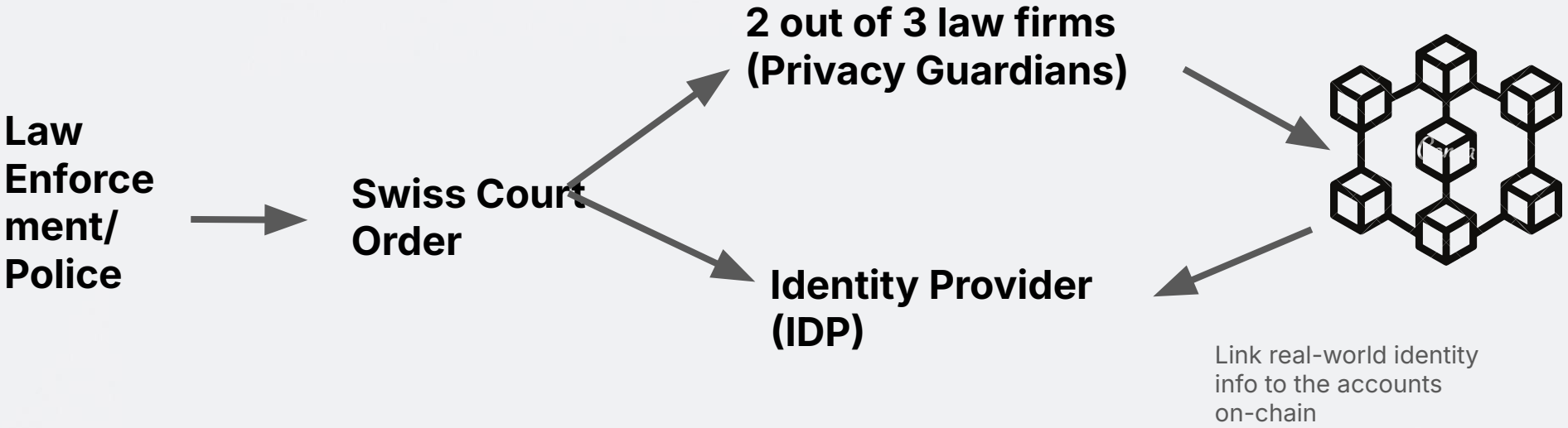
Wallet opens accounts on-chain



Wallet proves that it possesses signed identity info

Signed identity info stored in IDP database and wallet

Identity Disclosure Process



Principles of Privacy - Separation of Powers

No single party can link a user's Identity to the accounts they have on Concordium. This can only be done through the Identity disclosure process.

- IDPs (Identity Providers) cannot identify a user's account (or on-chain wallet address). They cannot connect an identity to on-chain activity or an address.
- A single Privacy Guardian cannot decrypt a user's Identity Disclosure Data. This means they cannot access the mapping between a user's identity and their on-chain presence. In addition they do not have access to the PII which is stored within the IDP's systems.
- It is only possible to connect a user's PII identity to an account by following the identity disclosure process.

Principles of Privacy - Users are Empowered through Zero-knowledge (ZK) Proofs

With verified identities in the wallets, users can prove:
- Their name is xyz (selectively disclose some info of their identity).
- They are older/younger than xyz (range proofs):
e.g., proving they are over 18 years without revealing their birth date.
- They belong or do not belong to a group (set/not-set membership proofs):
e.g., proving they are a citizen of the EU





Demo Time

Proof Explorer

ZK Proof Explorer Deep Dive

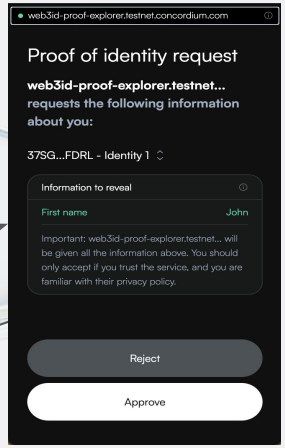




1. Send Request

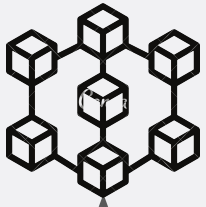
2. Return ZK Proof

Frontend



Wallet

Blockchain



5. Look Up Cryptographic Commitment

3. Send ZK Proof

5. Proof Valid / not Valid



Backend Verifier

Is The ZK Part Hard for Devs?

- No need to build a ZK dApp (plain PLTs, smart contracts dApps exist)

- If you want to use ZK, no ZK moon math knowledge needed by devs because:

    - Wallets do all the ZK proof generation:
    e.g., call the wallet API endpoint `generate_proof`.

    - ZK verifier backend provided by Concordium
    for rapid prototyping and testing:
    e.g., call the backend API endpoint `verify_proof`.



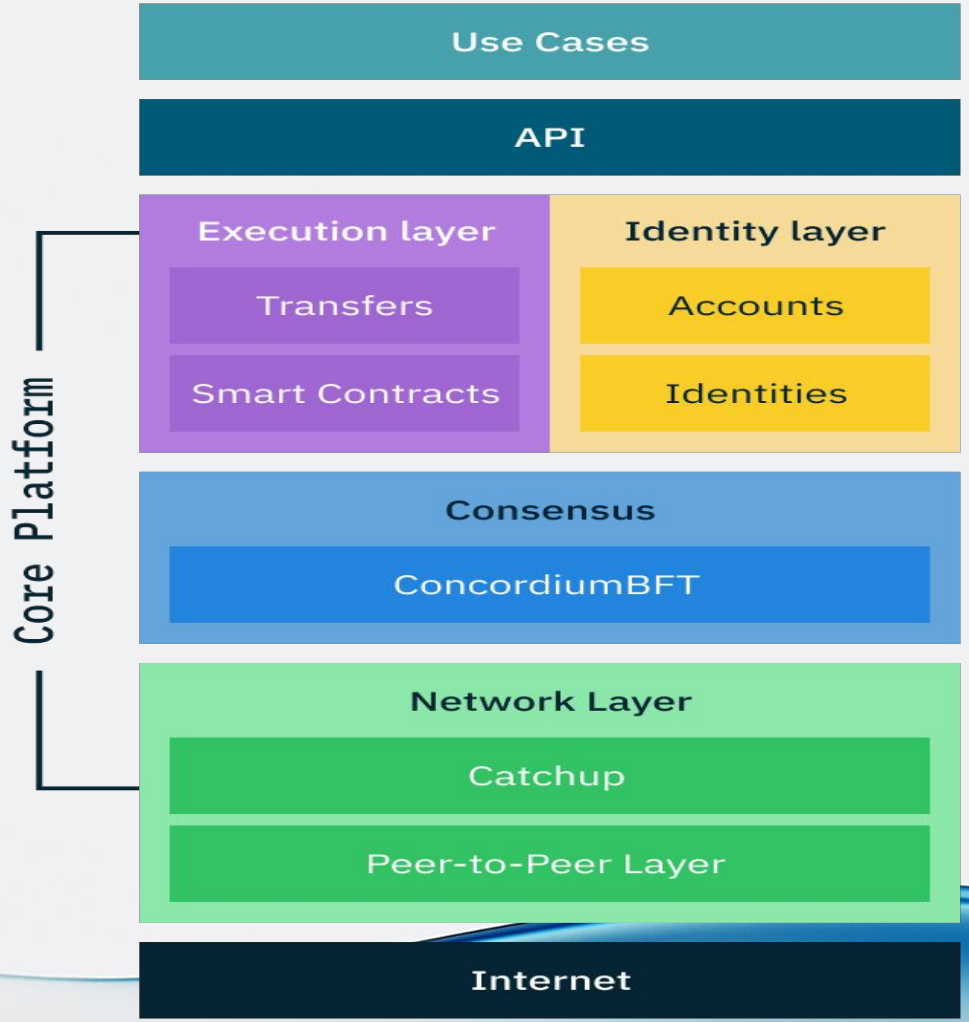
Architecture Overview

Concordium Core Features

- Identity layer
- Fast and final transactions (after 2-4 seconds)
- Predictable transaction costs - stable in Euros
- Protocol-Level Tokens

Protocol Architecture

- Network layer
  - The network layer handles communication between nodes.
- Consensus layer
  - The consensus layer ensures all nodes agree on the state of the blockchain using ConcordiumBFT, a novel proof-of-stake Byzantine fault tolerance consensus protocol.
- Identity layer
  - The identity layer manages user verification and privacy. It works with external identity providers who verify users' real-world identities. This information is stored securely off-chain, while on-chain transactions maintain user privacy.
- Execution layer
  - The execution layer allows users to interact with the platform through its API, processing transactions and running smart contracts. It uses WebAssembly as its foundation.


Use Cases
API
Execution layer
Transfers
Smart Contracts
Identity layer
Accounts
Identities
Consensus
ConcordiumBFT
Network Layer
Catchup
Peer-to-Peer Layer
Internet
Core Platform



Protocol-Level Tokens

PLTs

Protocol-Level Tokens (PLTs) are tokens issued and managed directly on the Concordium blockchain without a need for smart contracts. PLTs provide enhanced security by reducing attack vectors, improved efficiency in execution, and direct governance controls for token issuers.

Concordium has onboarded 11 stablecoin issuers in just the last 5 months.

Concordium's Protocol-Level Tokens (PLTs) deliver secure, compliant, and efficient programmable assets built for today's PayFi economy — smart money, not just smart contracts.

PLT Operations

Follow this link for code examples that showcase the following operations in Typescript and Rust:

- Mint tokens: Mint new tokens (issuer only)
- Burn tokens: Burn existing tokens (issuer only)
- Add account to allow list: Add an account to the allow list for a PLT (issuer only)
- Remove account from allow list: Remove an account from the allow list for a PLT (issuer only)
- Add account to deny list: Add an account to the deny list for a PLT (issuer only)
- Remove account from deny list: Remove an account from the deny list for a PLT (issuer only)
- Pause: Suspends balance transfer operations for a PLT (issuer only)
- Unpause: Resumes balance transfer operations for a PLT (issuer only)





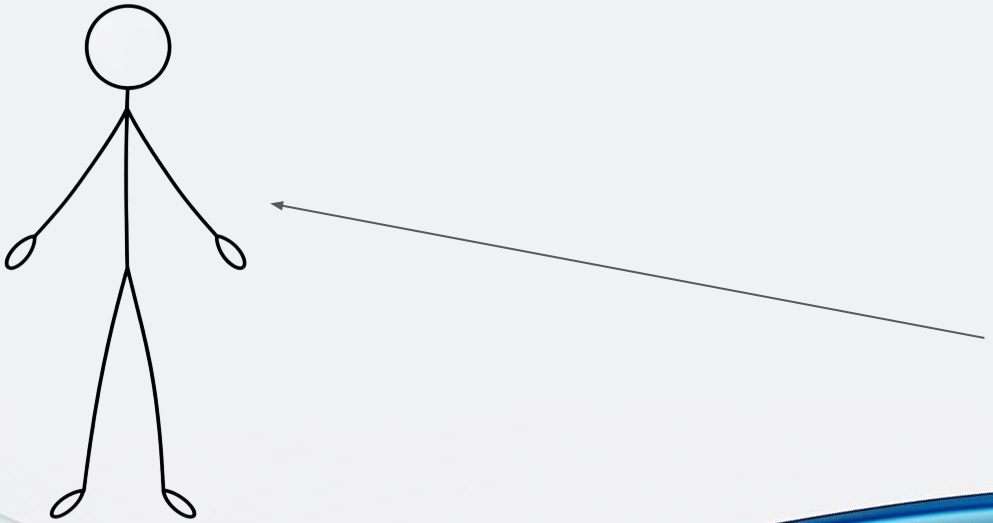
Installing Prerequisites

Set up your development environment by following this guide

Developer Tools

- Concordium Wallet for Web (Browser Wallet)
- Concordium Client (Command Line Tool)
- CCDScan (Blockexplorer)
- Public testnet node gRPC address: grpc.testnet.concordium.com
- SDKs and APIs
- Proof Explorer
- At the hackathon, there will be a Docker Compose based local chain

A Very Realistic Problem



This is you

The Problem

Many platforms require you to upload your ID or passport to prove nationality, exposing all your personal data.

What if there was a better way?

What if you could prove you're an EU national without revealing your identity, address, or any sensitive information?



Enter: The Allowlist Faucet

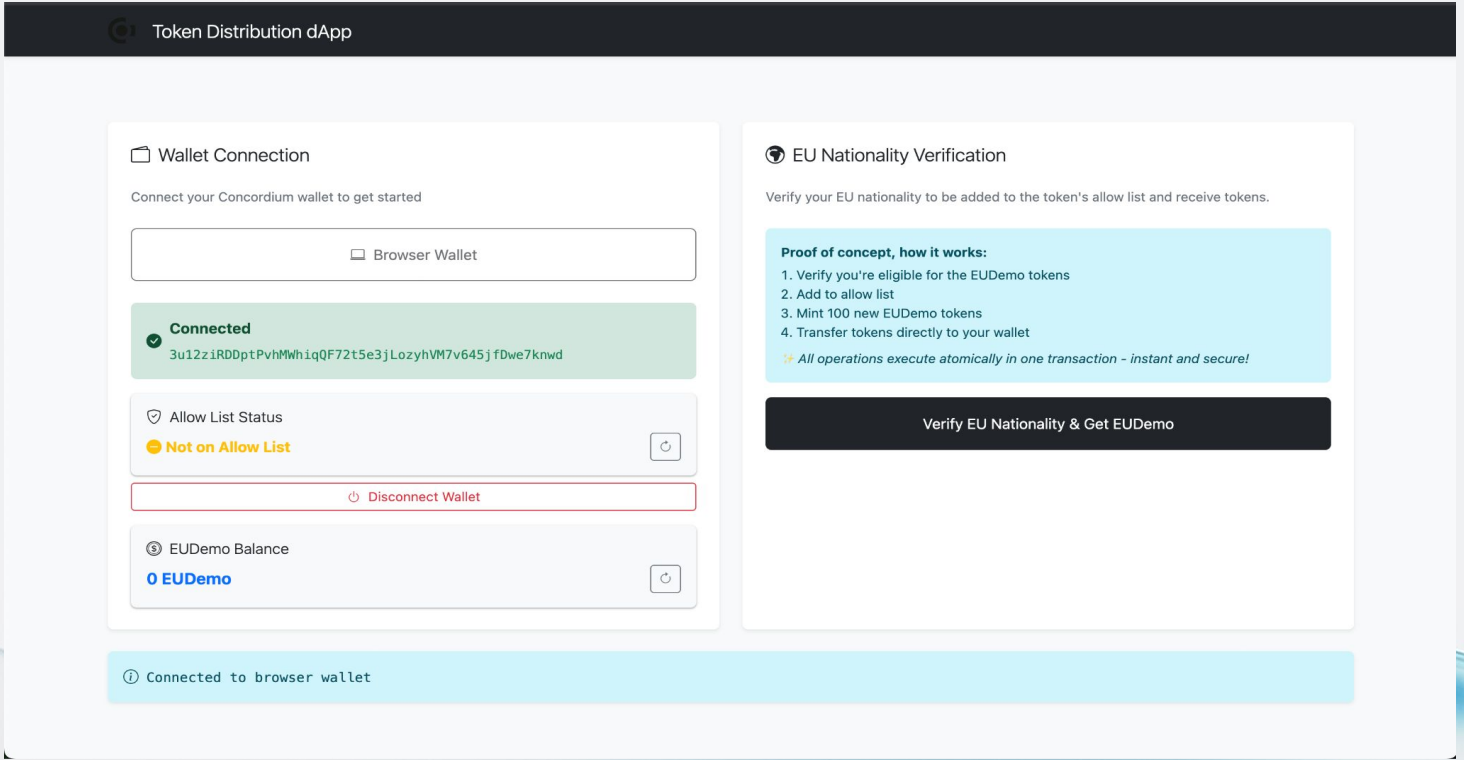
This demo app shows how Concordium's Protocol-Level Tokens (PLTs) protect your privacy by design.

How It Works

Instead of showing your ID, you generate a zero-knowledge proof that confirms only one thing: you're an EU national. Then, you’re eligible to receive tokens.

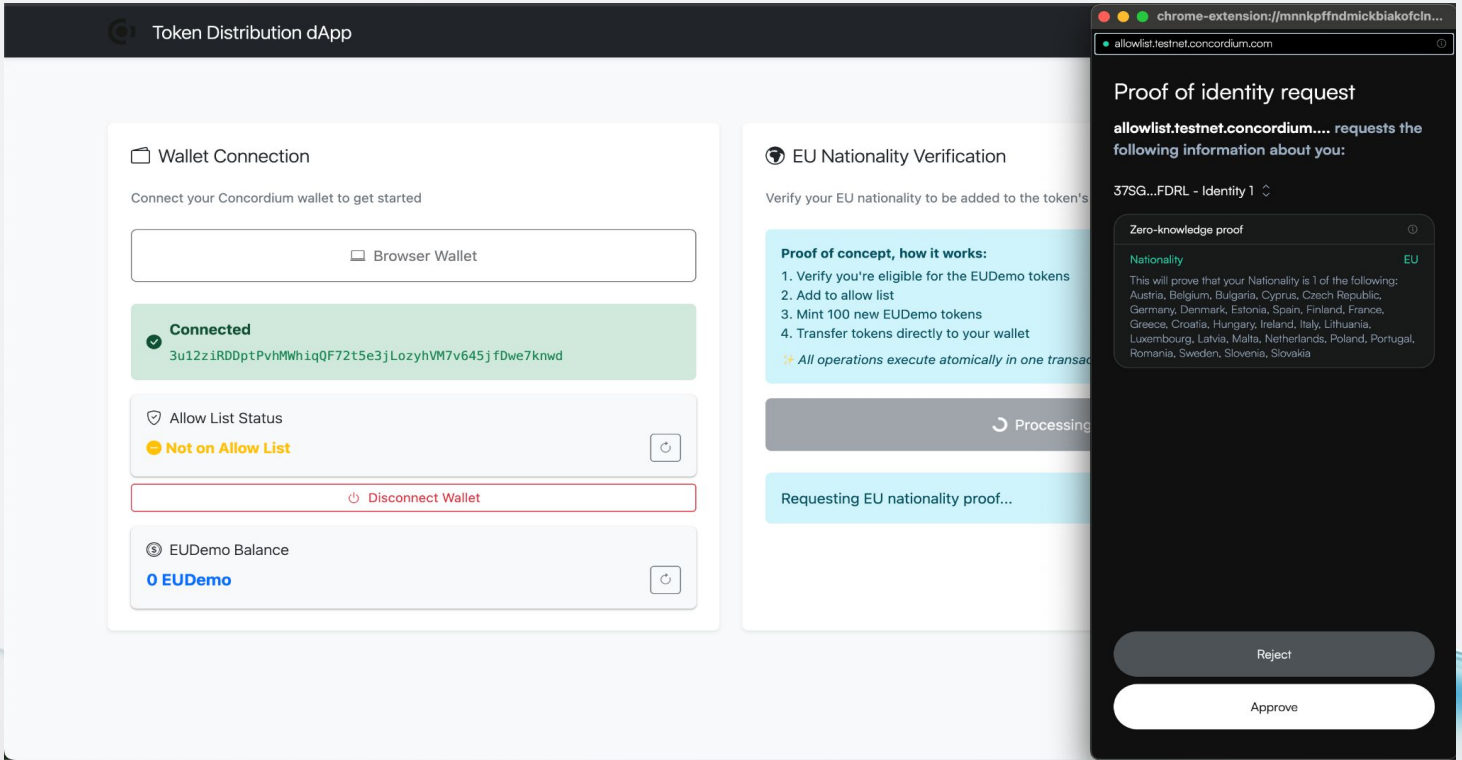
Opening the dApp

You connect your Concordium wallet to the Allowlist Faucet on testnet.



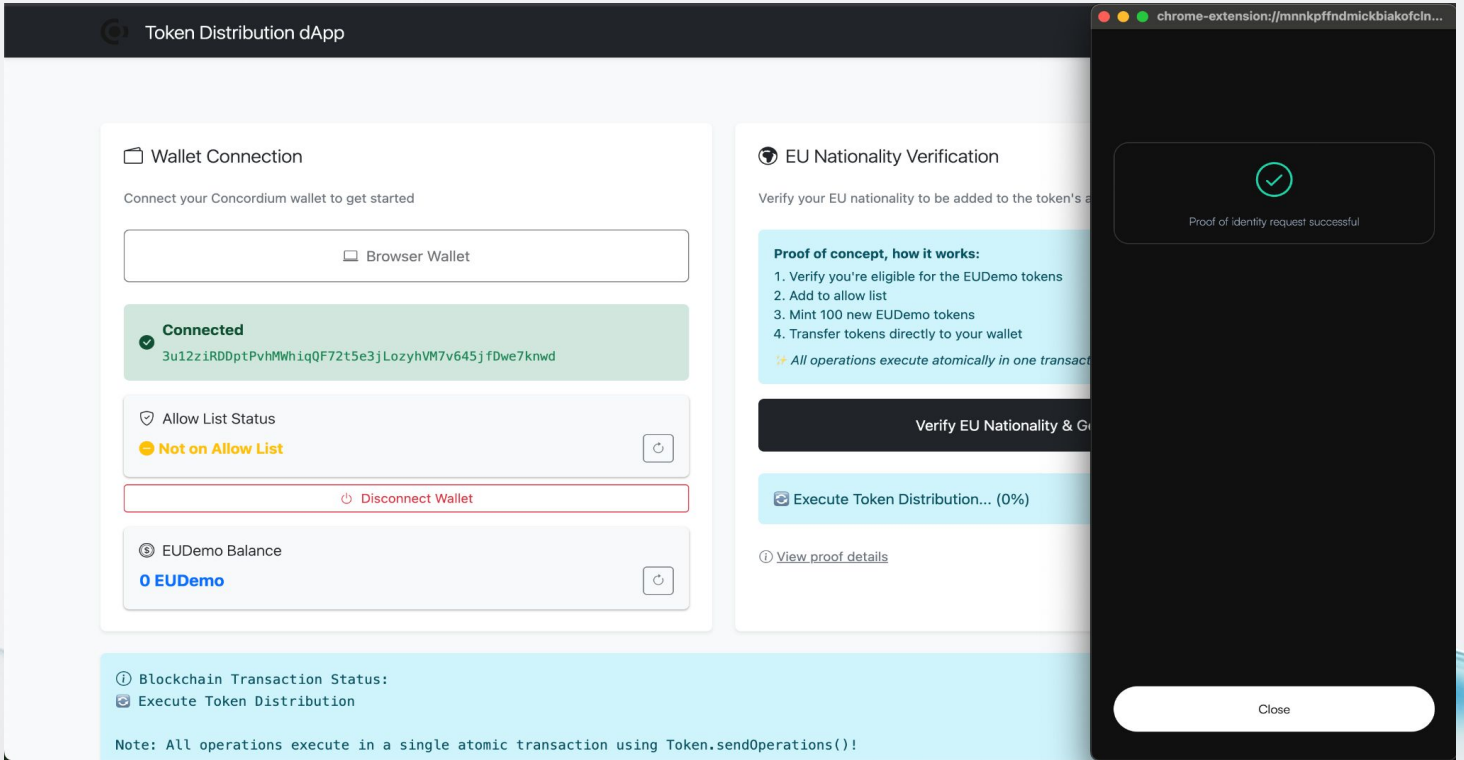
Generate Your Proof

With one click, your wallet creates a ZK proof from your identity—no documents, no data sharing.



Submit and Verify

The dApp verifies your proof off-chain, confirming you're eligible without ever seeing who you are.



Atomic Batch Transaction Begins

Once verified, the dApp executes three operations in a single atomic transaction. No need for smart contracts, PLTs operate directly at the protocol layer.

Step 1

Your account is added to the allow list for the EUDemo PLT

Step 2

100 Tokens are minted

Step 3

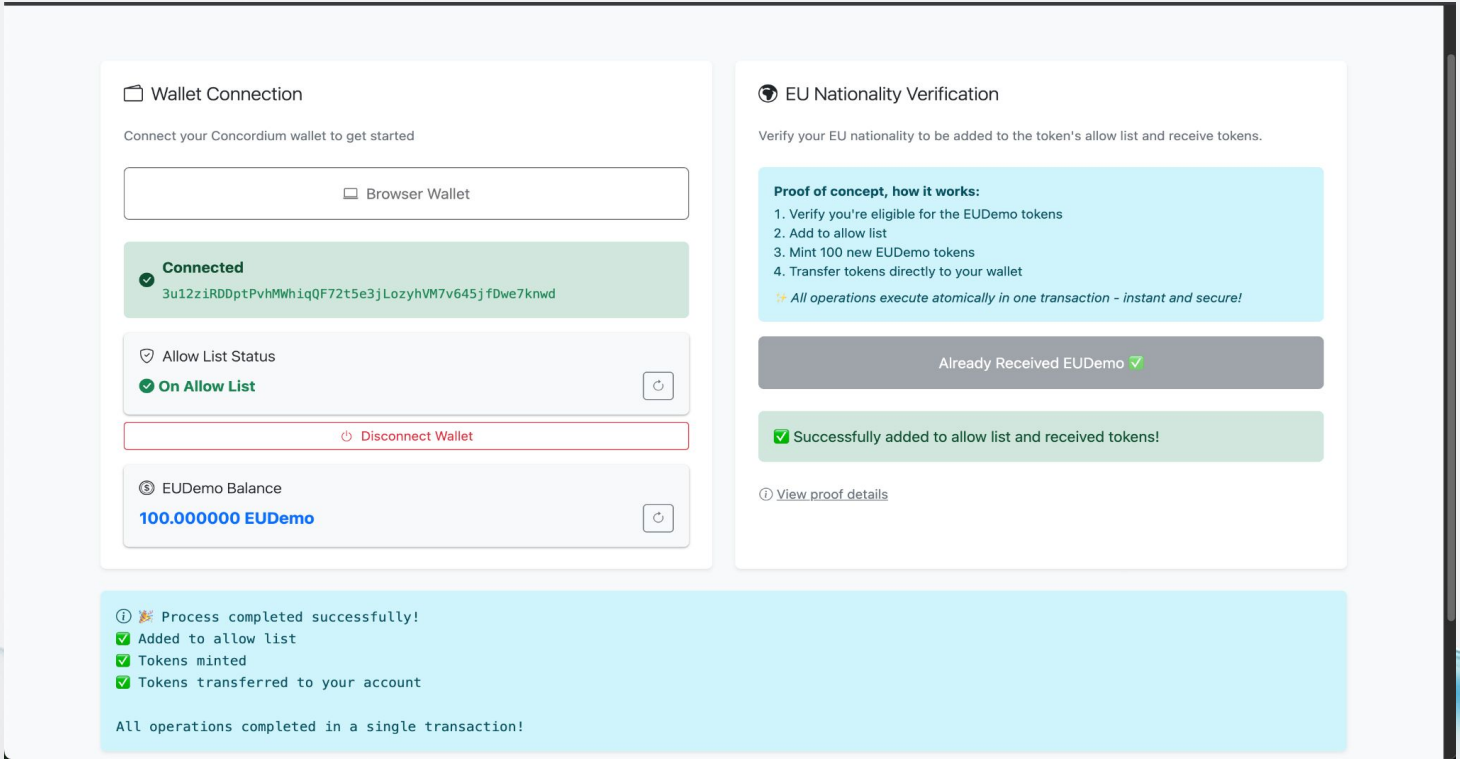
Tokens are transferred to your account

All or Nothing

The batch transaction is atomic, meaning that if any step fails, the entire transaction reverts—you're either fully verified with 100 tokens in your account, or nothing happens at all.

Your Privacy Intact

You now have 100 PLTs to spend, and your personal information never left your wallet.



Under the Hood: Technical Articles





Questions

Find us on Discord!

We will be here for help during the hackathon.

https://github.com/DOBEN
www.linkedin.com/in/DorisBenda/

https://github.com/dragosgd
https://www.linkedin.com/in/dragos-gabriel-danciulescu-087552213/

Thank you for your attention!